LAB4 : Facial Detection Systems

Part 2: Facial Detection using Deep Learning

Face detection is a fundamental problem in computer vision and is widely used in applications such as security systems, human–computer interaction, smart surveillance, and biometric authentication. In this assignment, students will implement a deep learning–based facial detection system using a Convolutional Neural Network (CNN).

The goal of this is to understand how neural networks can be trained to classify images as **face** or **non-face**. Students will explore the dataset, train a classifier, evaluate its performance, and visualize the predictions.

students will also briefly observe how machine learning systems may behave differently across different datasets or demographic groups. Understanding such behavior is important for designing responsible and ethical AI systems.

Learning Objectives

After completing this asssignemnt, students should be able to:

- Understand the basic workflow of a computer vision pipeline for facial detection.
- Implement and train a CNN-based binary image classifier. CLO2
- Evaluate the performance of a trained model on test data.
- Visualize and interpret the predictions of a deep learning model.

```
# Import Tensorflow 2.0
import tensorflow as tf

import IPython
import functools
import matplotlib.pyplot as plt
import numpy as np
from tqdm import tqdm

# # Download and import the MIT 6.S191 package
!pip install mitdeeplearning --quiet
import mitdeeplearning as mdl
```

2.1 Datasets

We'll be using three datasets in this lab. In order to train our facial detection models, we'll need a dataset of positive examples (i.e., of faces) and a dataset of negative examples (i.e., of things that are not faces). We'll use these data to train our models to classify images as either faces or not faces. Finally, we'll need a test dataset of face images.

1. **Positive training data**: CelebA Dataset. A large-scale (over 200K images) of celebrity faces.
2. **Negative training data**: ImageNet. Many images across many different categories. We'll take negative examples from a variety of non-human categories. Fitzpatrick Scale skin type classification system, with each image labeled as "Lighter" or "Darker".

Let's begin by importing these datasets. We've written a class that does a bit of data pre-processing to import the training data in a usable format.

```
# Get the training data: both images from CelebA and ImageNet
path_to_training_data = tf.keras.utils.get_file('train_face.h5', 'https://www.dropbox.com/s/hlz8atheyozp1yx/train_face.h5?dl=1'
# Instantiate a TrainingDatasetLoader using the downloaded dataset
loader = mdl.lab2.TrainingDatasetLoader(path_to_training_data)
```

```
Opening /root/.keras/datasets/train_face.h5
Loading data into memory...
```

We can look at the size of the training dataset and grab a batch of size 100:

```
number_of_training_examples = loader.get_train_size()
print(number_of_training_examples)
(images, labels) = loader.get_batch(100)
```

```
print(images.shape)
print(labels.shape)
```

```
109914
(100, 64, 64, 3)
(100, 1)
```

Play around with displaying images to get a sense of what the training data actually looks like!

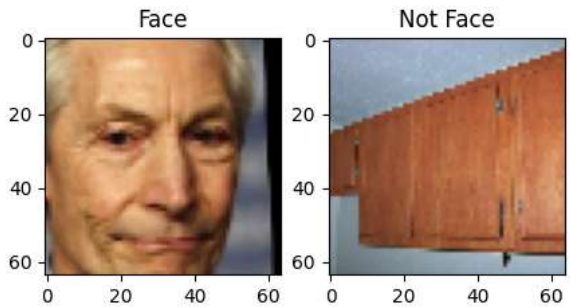
Change the sliders to look at positive and negative training examples!

```
### Examining the CelebA training dataset ###

#@title Change the sliders to look at positive and negative

face_images = images[np.where(labels==1)[0]]
not_face_images = images[np.where(labels==0)[0]]

idx_face = 24 #@param {type:"slider", min:0, max:50, step:1}
idx_not_face = 17 #@param {type:"slider", min:0, max:50, ste

plt.figure(figsize=(5,5))
plt.subplot(1, 2, 1)
plt.imshow(face_images[idx_face])
plt.title("Face"); plt.grid(False)

plt.subplot(1, 2, 2)
plt.imshow(not_face_images[idx_not_face])
plt.title("Not Face"); plt.grid(False)
```

idx_face 24

idx_not_face 17

Hide code



2.2 CNN for facial detection

we'll define and train a CNN on the facial classification task, and evaluate its accuracy. The CNN model has a relatively standard architecture consisting of a series of convolutional layers with batch normalization followed by two fully connected layers to flatten the convolution output and generate a class prediction.

The classifier should contain the following layers:

Layer	Type	Filters / Units	Kernel Size	Stride	Activation
1	Convolution	12	5×5	2	ReLU
2	Batch Normalization	—	—	—	—
3	Convolution	24	5×5	2	ReLU
4	Batch Normalization	—	—	—	—
5	Convolution	48	3×3	2	ReLU
6	Batch Normalization	—	—	—	—
7	Convolution	72	3×3	2	ReLU
8	Batch Normalization	—	—	—	—
9	Flatten	—	—	—	—
10	Dense	512	—	—	ReLU
11	Output Layer	1	—	—	Linear

```
### Define the CNN model ###
n_filters = 12

# Build the CNN classifier
standard_classifier = tf.keras.Sequential([
```

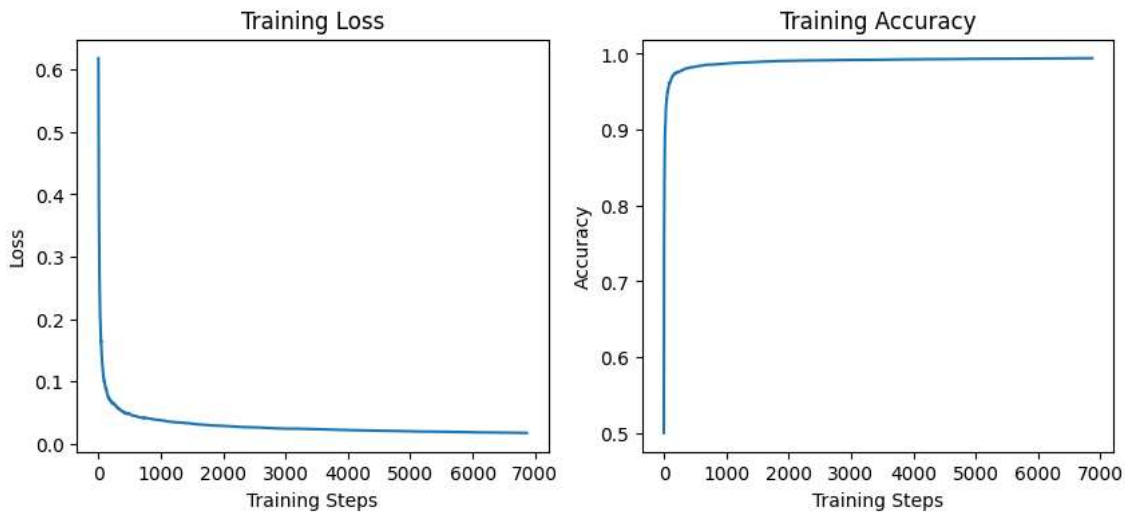
```
    # 1. Convolutional Layer
    # Note: We specify the input_shape here. The MIT CelebA dataset crops are typically 64x64 RGB images.
    tf.keras.layers.Conv2D(n_filters, (5, 5), strides=2, activation='relu', input_shape=(64, 64, 3)),

    # 2. Batch Normalization
    tf.keras.layers.BatchNormalization(),

    # 3. Convolutional Layer (n_filters * 2 = 24)
    tf.keras.layers.Conv2D(n_filters * 2, (5, 5), strides=2, activation='relu'),

    # 4. Batch Normalization
    tf.keras.layers.BatchNormalization(),

    # 5. Convolutional Layer (n_filters * 4 = 48)
    tf.keras.layers.Conv2D(n_filters * 4, (3, 3), strides=2, activation='relu'),

    # 6. Batch Normalization
    tf.keras.layers.BatchNormalization(),

    # 7. Convolutional Layer (n_filters * 6 = 72)
    tf.keras.layers.Conv2D(n_filters * 6, (3, 3), strides=2, activation='relu'),

    # 8. Batch Normalization
    tf.keras.layers.BatchNormalization(),

    # 9. Flatten
    tf.keras.layers.Flatten(),

    # 10. Dense Layer
    tf.keras.layers.Dense(512, activation='relu'),

    # 11. Output Layer (1 node for binary classification: Face vs. Not Face)
    tf.keras.layers.Dense(1, activation='linear')
])

# Optional: Print the summary to verify the architecture
# standard_classifier.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shap
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Now let's train the standard CNN!

```
# Training hyperparameters
params = dict(
  batch_size = 32,
  num_epochs = 2,
  learning_rate = 5e-4,
)

# TODO 2: Define optimizer using Adam
optimizer = tf.keras.optimizers.Adam(learning_rate=params["learning_rate"])

# TODO 3: Compile the model
standard_classifier.compile(
    optimizer=optimizer,
    loss=tf.keras.losses.BinaryCrossentropy(from_logits=True),
    metrics=['accuracy']
)

# Train the model using batches from the dataset loader
loss_history = []
acc_history = []

for epoch in range(params["num_epochs"]):

    # Calculate the number of steps per epoch
    for step in range(loader.get_train_size() // params["batch_size"]):

        # Get a batch of images and labels
        images, labels = loader.get_batch(params["batch_size"])

        # TODO 4: train the model using train_on_batch
        # train_on_batch performs a single gradient update on a single batch of data
        loss, acc = standard_classifier.train_on_batch(images, labels)
```

```
        # TODO 5: store loss and accuracy values in loss_history and acc_history
        loss_history.append(loss)
        acc_history.append(acc)

# Visualize the training performance
# Plot loss and accuracy curves using matplotlib
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
plt.plot(loss_history)
plt.title("Training Loss")
plt.xlabel("Training Steps")
plt.ylabel("Loss")

plt.subplot(1,2,2)
plt.plot(acc_history)
plt.title("Training Accuracy")
plt.xlabel("Training Steps")
plt.ylabel("Accuracy")

plt.show()
```



Evaluate performance of the standard CNN

Next, let's evaluate the classification performance of our CelebA-trained standard CNN on the training dataset.

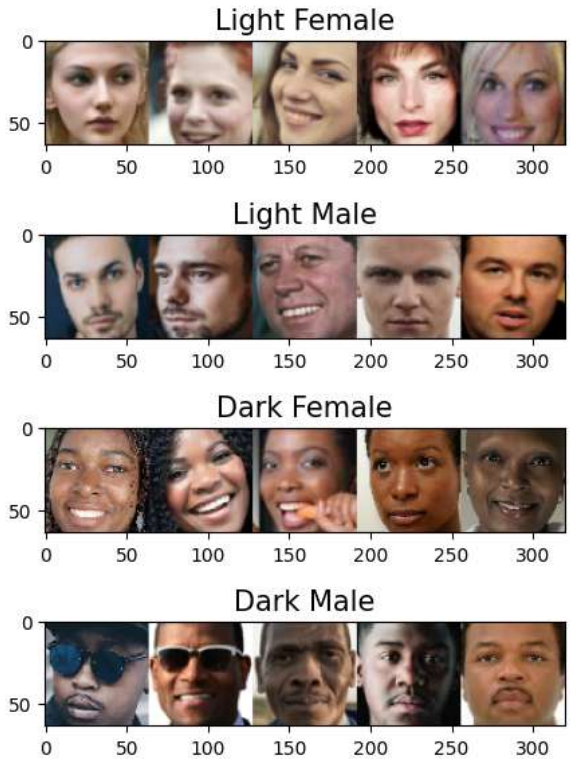
```
### Evaluate the CNN classifier ###

# TODO 6: Load a large batch of images and labels from the dataset
# A batch of 500 is a good sample size to test on
images, labels = loader.get_batch(500)

# TODO 7: Use the trained CNN model to generate predictions for the images
# This outputs the raw 'logits'
logits = standard_classifier.predict(images)

# TODO 8: Convert the logits to probabilities using the sigmoid function
# This squishes the raw numbers into a range between 0.0 and 1.0
probs = tf.sigmoid(logits)

# TODO 9: Convert probabilities into binary predictions (0 or 1)
# Anything >= 0.5 becomes 1 (Face), anything < 0.5 becomes 0 (Not Face)
preds = tf.round(probs)

# TODO 10: Compute the classification accuracy
# Compare our binary predictions to the true labels and find the average matches
correct_predictions = tf.equal(tf.cast(preds, tf.int32), tf.cast(labels, tf.int32))
acc_standard = tf.reduce_mean(tf.cast(correct_predictions, tf.float32))

# Print the final accuracy
print("Standard CNN accuracy on training set: {:.4f}".format(acc_standard.numpy()))
```

```
16/16 ━━━━━━━━━━━━━━━━━━━━ 1s 48ms/step
Standard CNN accuracy on training set: 1.0000
```

We will also evaluate our networks on an independent test dataset containing faces that were not seen during training. For the test data, we'll look at the classification accuracy across four different demographics, based on the Fitzpatrick skin scale and sex-based labels: dark-skinned male, dark-skinned female, light-skinned male, and light-skinned female.

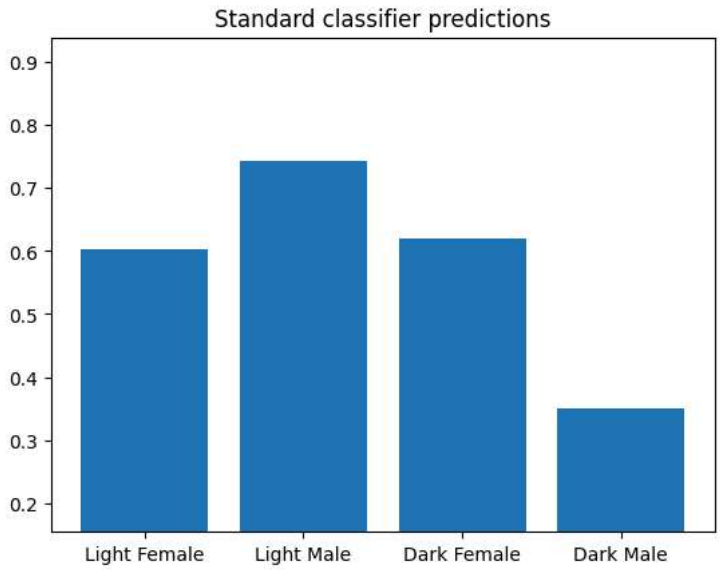
Let's take a look at some sample faces in the test set.

```
### Load test dataset and plot examples ###

test_faces = mdl.lab2.get_test_faces()
keys = ["Light Female", "Light Male", "Dark Female", "Dark Male"]
for group, key in zip(test_faces,keys):
  plt.figure(figsize=(5,5))
  plt.imshow(np.hstack(group))
  plt.title(key, fontsize=15)
```



Now, let's evaluate the probability of each of these face demographics being classified as a face using the standard CNN classifier we've just trained.

```
### Evaluate the standard CNN on the test data ###

standard_classifier_logits = [standard_classifier(np.array(x, dtype=np.float32)) for x in test_faces]
standard_classifier_probs = tf.squeeze(tf.sigmoid(standard_classifier_logits))

# Plot the prediction accuracies per demographic
xx = range(len(keys))
yy = standard_classifier_probs.numpy().mean(1)
plt.bar(xx, yy)
plt.xticks(xx, keys)
plt.ylim(max(0,yy.min()-np.ptp(yy)/2.), yy.max()+np.ptp(yy)/2.)
plt.title("Standard classifier predictions");
```




```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is depreca
  return datetime.utcnow().replace(tzinfo=utc)
```

Take a look at the accuracies for this first model across these four groups. What do you observe? Would you consider this model biased or unbiased? What are some reasons why a trained model may have biased accuracies?

The model exhibits significant algorithmic bias, as evidenced by its drastically uneven performance: it achieves its highest accuracy on Light Males (~74%) while performing exceptionally poorly on Dark Males with both female groups falling in the middle. This disparity is